Information Management 2 — Backend & Database System Rubric

Audience: 2nd-Year College Students | 5-Point Scale | Total: 100 Points

Criteria	5 – Excellent	4 – Very Good	3 – Satisfactory	2 – Developing	1 – Beginning
1. Database Connection & Backend Integration (20 pts)	Database connection is 100% successful; all required PHP backend functions communicate with the database correctly; 0 connection errors during testing.	Connection is successful in 90–99% of tests; 1 minor error occurs but does not prevent normal operation.	Connection is successful in 75–89% of tests; up to 2 errors occur.	Connection is successful in 50–74% of tests; 3–4 errors occur.	Connection is successful in below 50% of tests or the system does not connect to the database.
2. Normalization (1NF–3NF) (20 pts)	100% of applicable tables follow 1NF, 2NF, and 3NF with no major redundancy or dependency errors.	90–99% of tables are correctly normalized; 1 minor issue is present.	75–89% of tables are correctly normalized; 2–3 issues are present.	50–74% of tables are correctly normalized; 4–5 issues are present.	Below 50% of tables are properly normalized or normalization is not demonstrated.
3. CREATE Functionality (15 pts)	Successfully creates 100% of required test records and stores them correctly in the database.	Successfully creates 90–99% of test records.	Successfully creates 75–89% of test records.	Successfully creates 50–74% of test records.	Successfully creates below 50% of test records or function is not working.
4. READ Functionality (10 pts)	Retrieves and displays 100% of required database records accurately.	Retrieves 90–99% of required records accurately.	Retrieves 75–89% of required records.	Retrieves 50–74% of required records.	Retrieves below 50% of records or READ is not functional.
5. UPDATE Functionality (10 pts)	Correctly updates 100% of tested records and changes are reflected in the database.	Correctly updates 90–99% of tested records.	Correctly updates 75–89% of tested records.	Correctly updates 50–74% of tested records.	Correctly updates below 50% of tested records or UPDATE is not functional.
6. ARCHIVE Functionality (10 pts)	Successfully archives 100% of tested records without permanently deleting the original database record.	Successfully archives 90–99% of tested records.	Successfully archives 75–89% of tested records.	Successfully archives 50–74% of tested records.	Successfully archives below 50% of tested records or ARCHIVE is not functional.
7. ERD, Relationships & Crow's Foot Implementation (15 pts)	100% of implemented tables and relationships match the logical ERD; PKs, FKs, cardinalities, and Crow's Foot notation are correctly implemented.	90–99% match the ERD with 1 minor relationship/key error.	75–89% match the ERD with 2–3 errors.	50–74% match the ERD with 4–5 errors.	Below 50% match the ERD or relationships are largely missing/incorrect.

Point Conversion

Criterion Score = (Level ÷ 5) × Criterion Points. Example: 4/5 for Database Connection = 4 ÷ 5 × 20 = 16 points.

Minimum Backend Requirement

- PHP successfully connects to the actual database.
- CREATE, READ, UPDATE, and ARCHIVE are functional.
- Actual database records are stored and retrieved; no hard-coded/sample data as the main implementation.
- Database applies 1NF, 2NF, and 3NF.
- Primary Keys and Foreign Keys are correctly implemented.
- Relationships match the Logical ERD and use Crow's Foot notation.

Assessment Focus: The rubric measures the student's ability to apply Information Management 2 concepts through a working backend and database system. A visually attractive interface without a successful database connection or functional backend should not receive a high score.